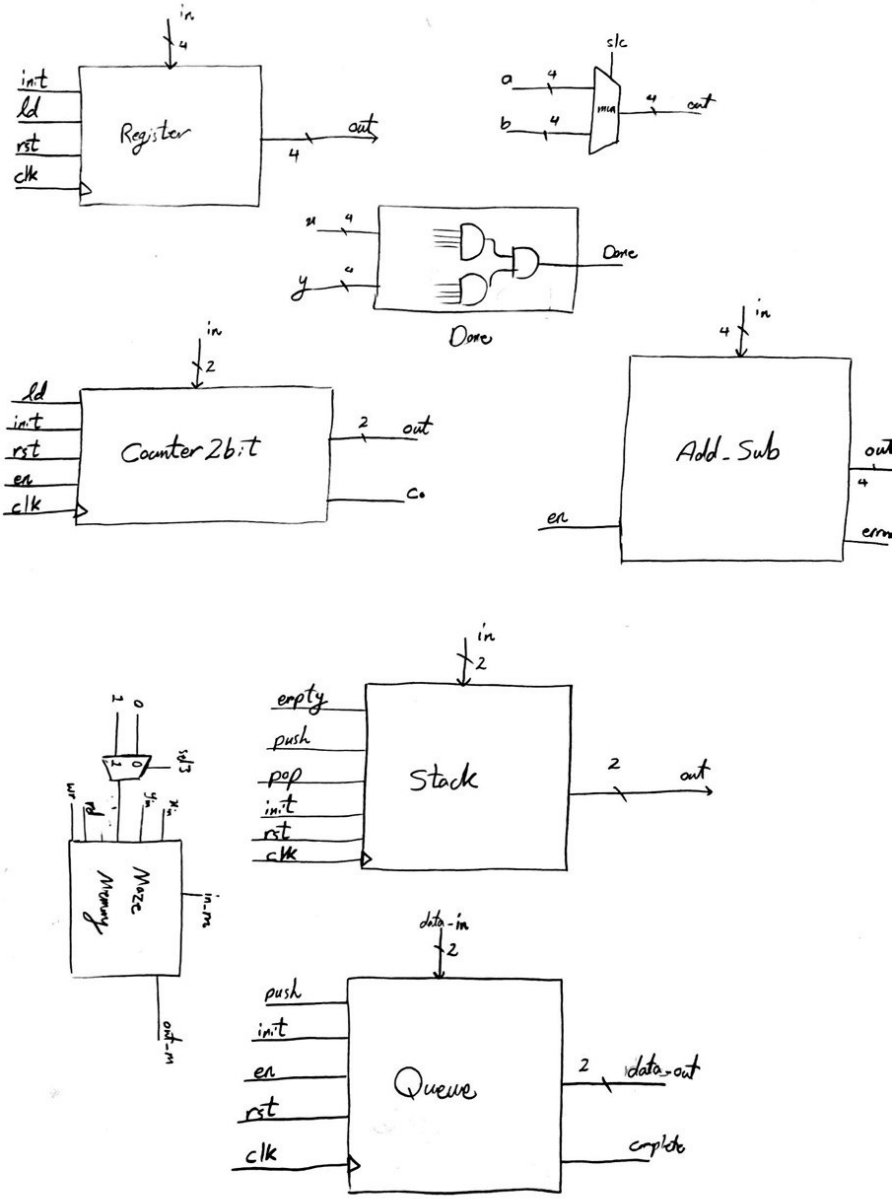
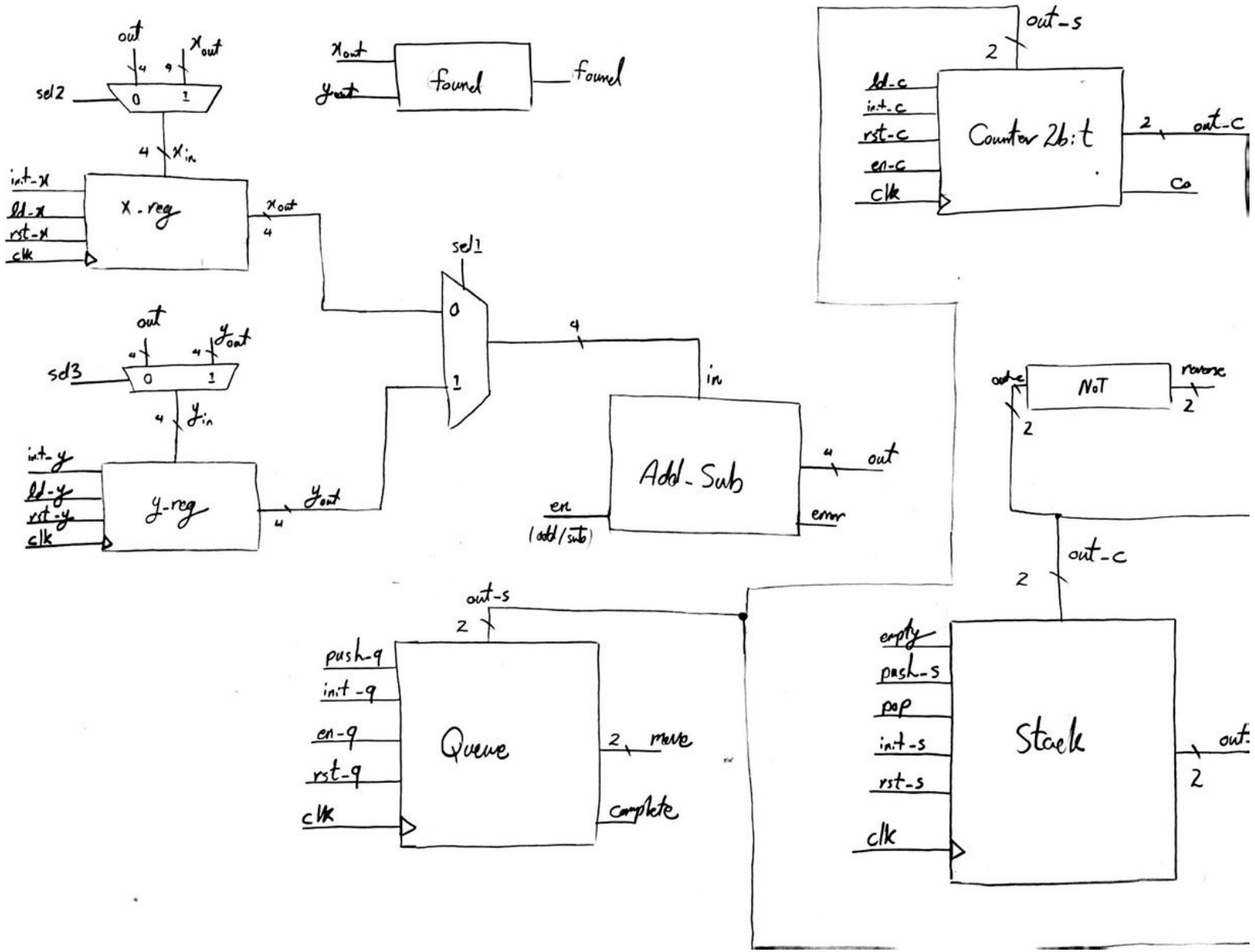
# CA1_Maze

Ali Mahdian
810801079

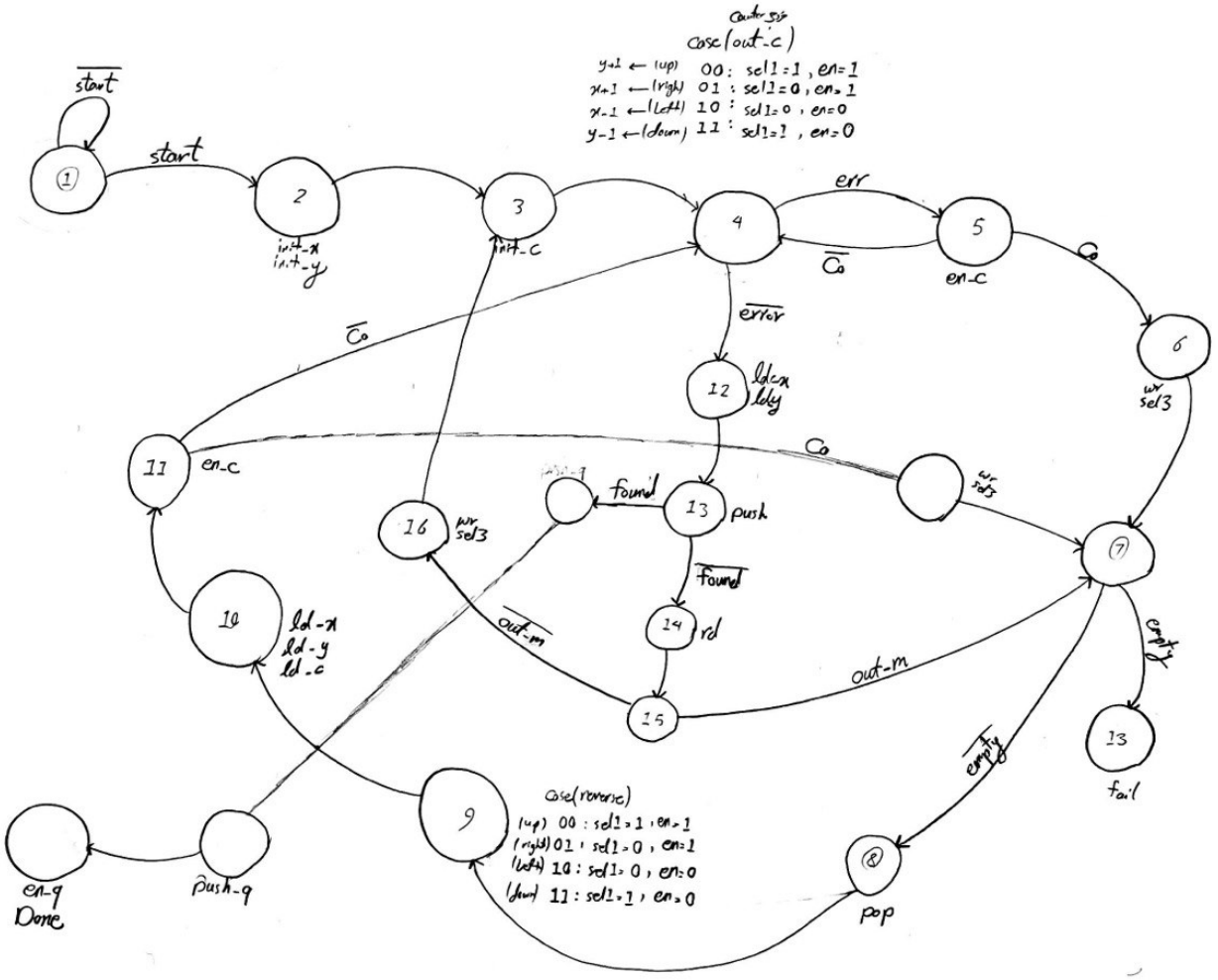
Deniz Mostafazadegan
810102603

Data Path :

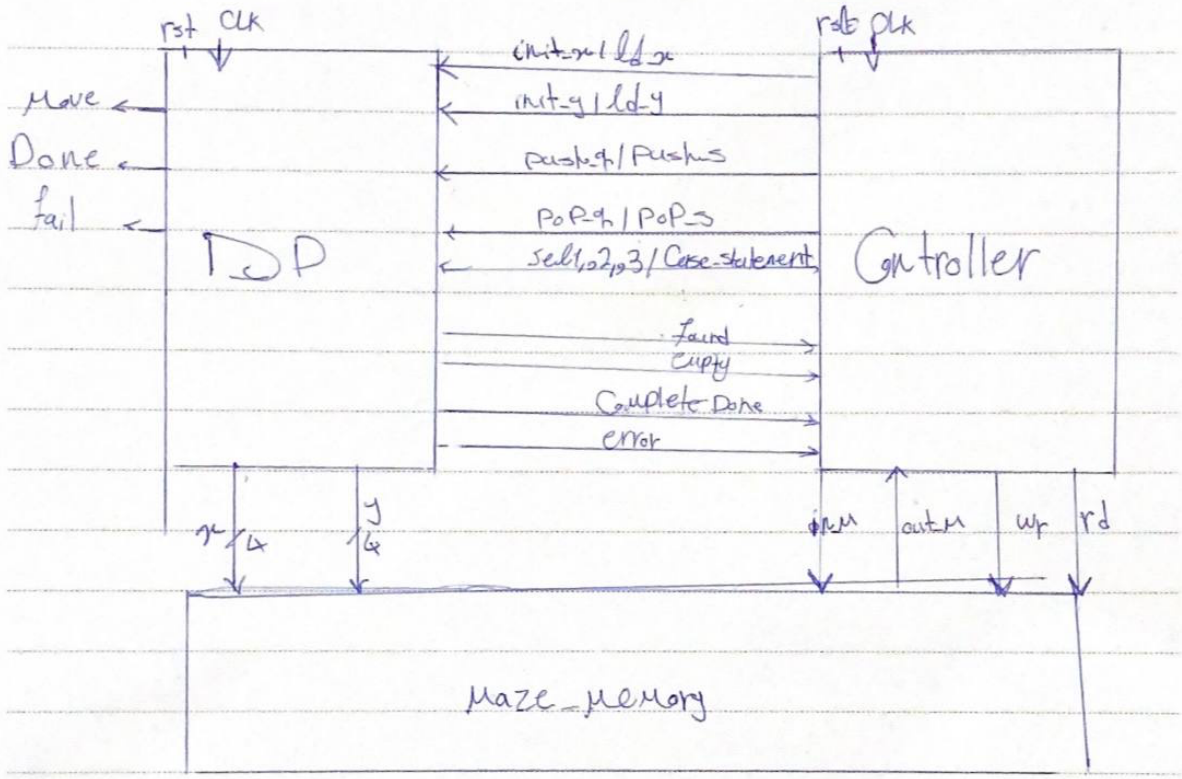



out
x_out
sel2
0
1
4
x_in
init_x
ld_x
rst_x
clk
X_reg
x_out
4
x_out
y_out
found
found
out
y_out
sel3
y_in
init_y
ld_y
rst_y
clk
y_reg
y_out
sel1
in
Add_Sub
en
(add/sub)
out
error
out_s
2
ld_c
init_c
rst_c
en_c
clk
Counter 2bit
out_c
Co
NOT
reverse
out_c
push_q
init_q
en_q
rst_q
clk
Queue
move
complete
empty
push_s
pop
init_s
rst_s
clk
Stack
out

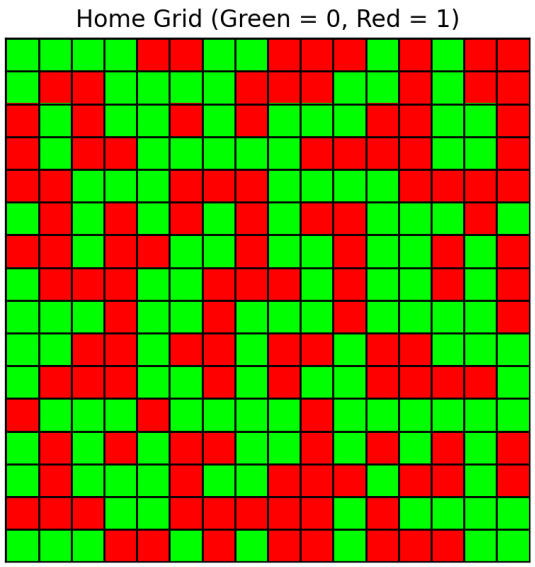
Controller :


counter_sig
case(out_c)
y+1 ← (up) 00: sel1=1, en=1
x+1 ← (right) 01: sel1=0, en=1
x-1 ← (left) 10: sel1=0, en=0
y-1 ← (down) 11: sel1=1, en=0
start
start
err
error
found
Found
out_m
empty
case(reverse)
(up) 00: sel1=1, en=1
(right) 01: sel1=0, en=1
(left) 10: sel1=0, en=0
(down) 11: sel1=1, en=0
init_x
init_y
init_c
en_c
wr sel3
ld_x
ld_y
ld_c
push
rd
pop
en_q
Done
push_q
fail

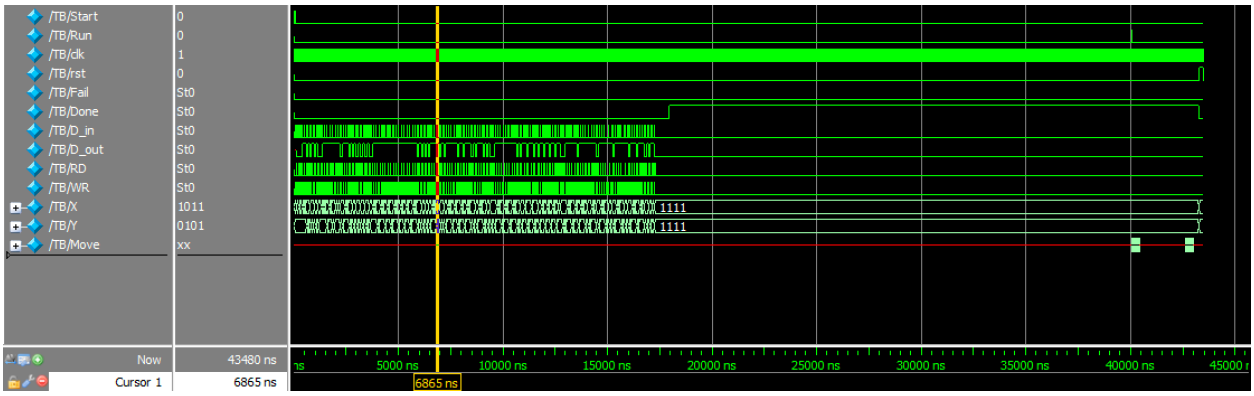
Date


rst
CLK
rst
clk
init_x / ld_x
init_y / ld_y
push_s / Pushs
PoP_s / PoP_s
sel1,2,3 / Case_statement
move
Done
fail
DP
Controller
found
empty
Complete Done
error
x
y
addr
data_in
wr
rd
maze_memory

Example 1)

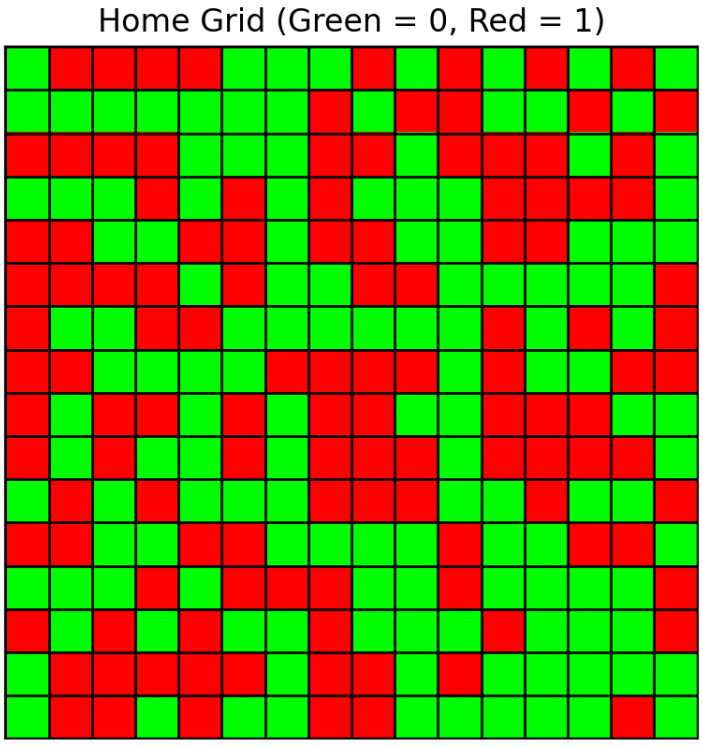



```
right
right
right
down
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right
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left
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right
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down
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right
down
```

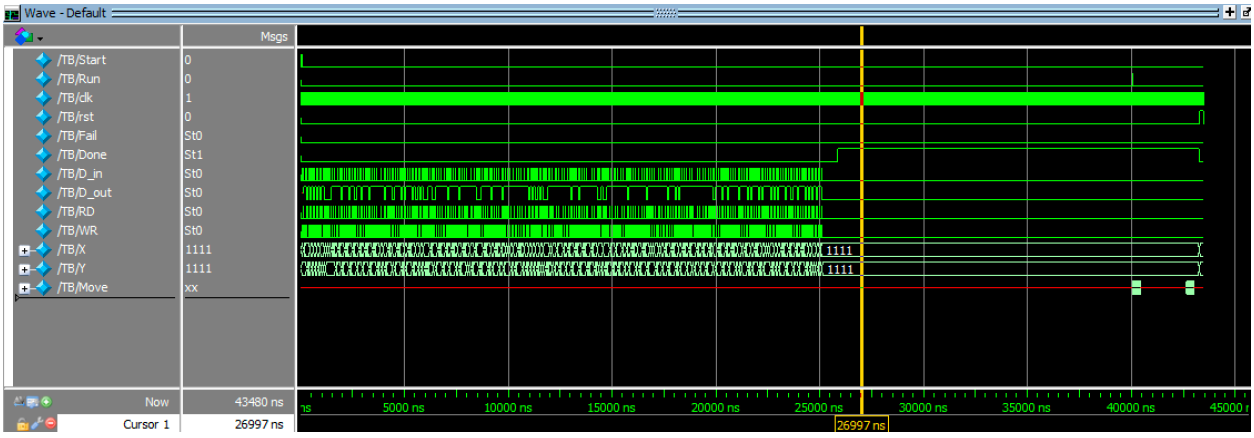



The Done = 1 so there's way !

Example 2)

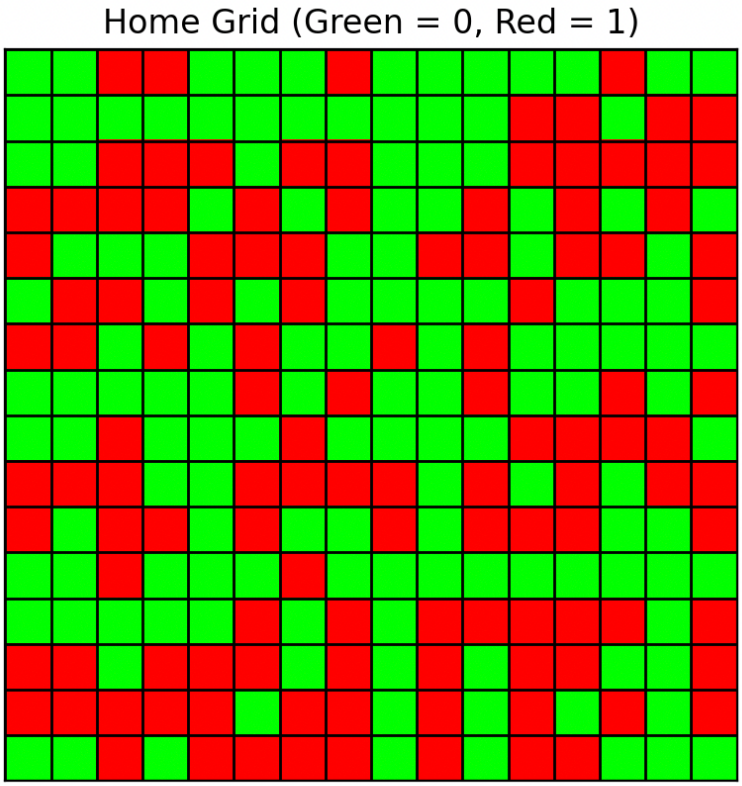


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left
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right
right
right
down
```

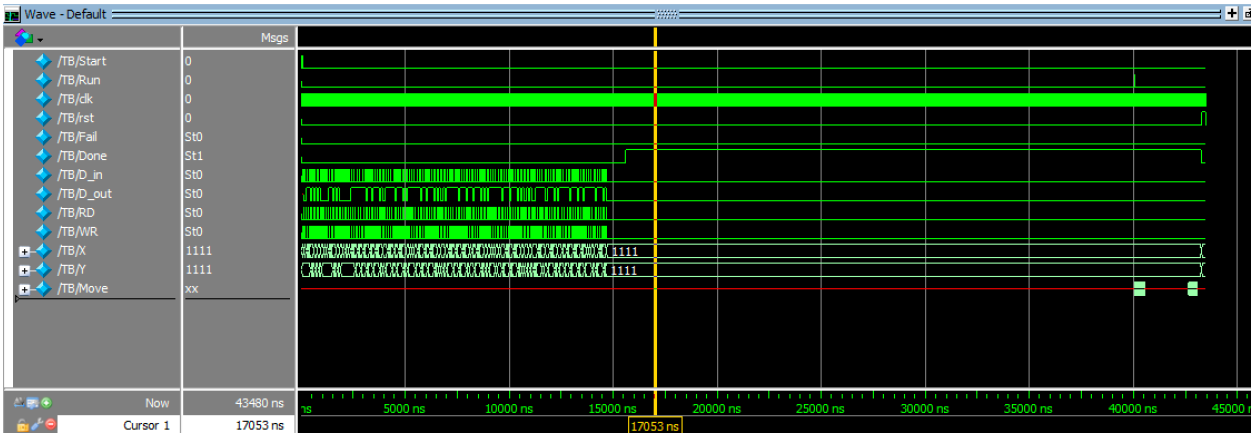



The Done = 1 so there’s way !

Example 3)

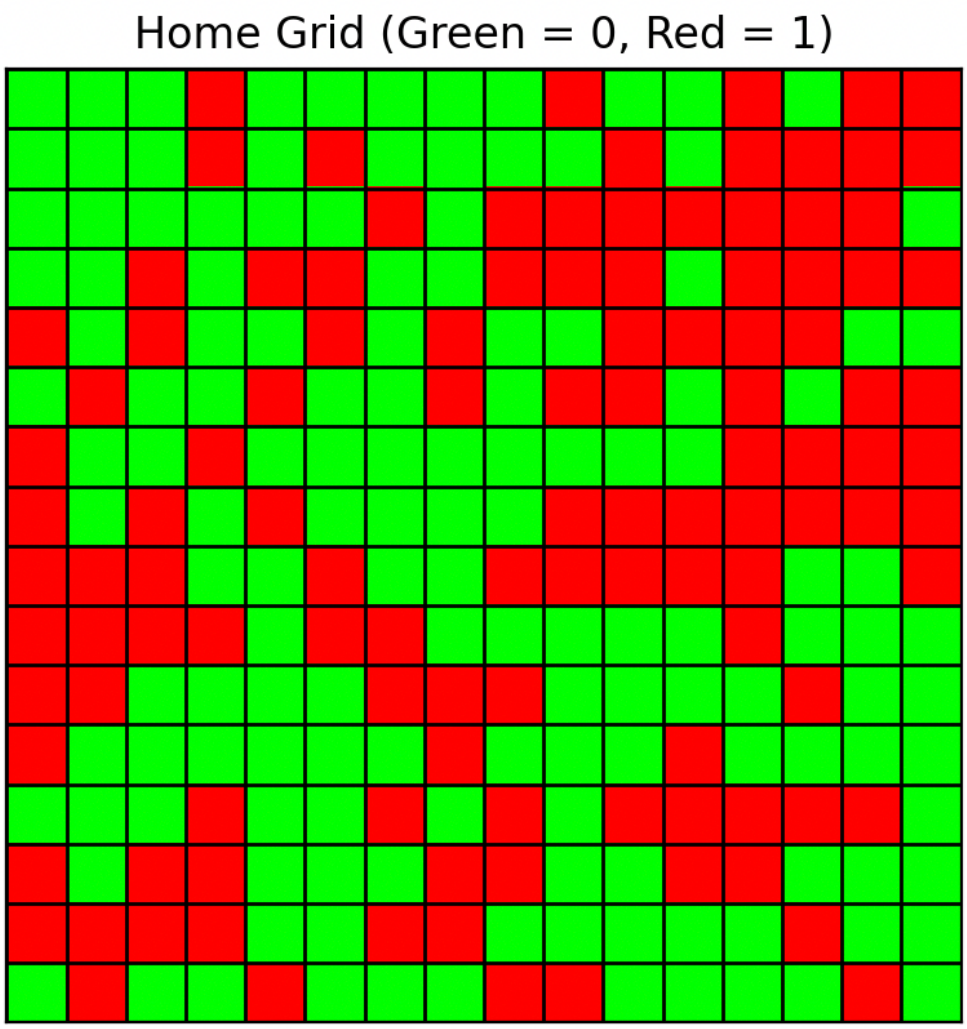


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right
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down
down
right
```

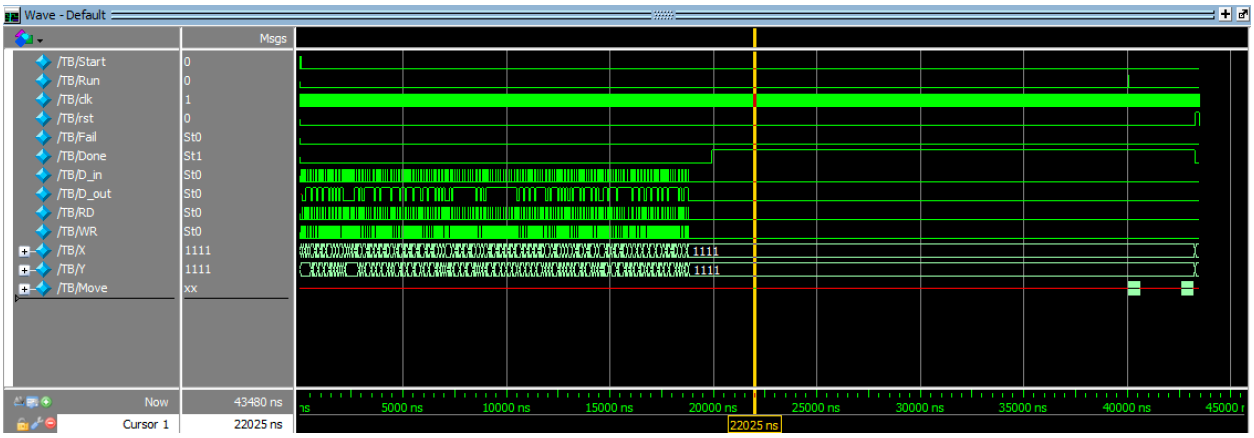



The Done = 1 so there’s way !

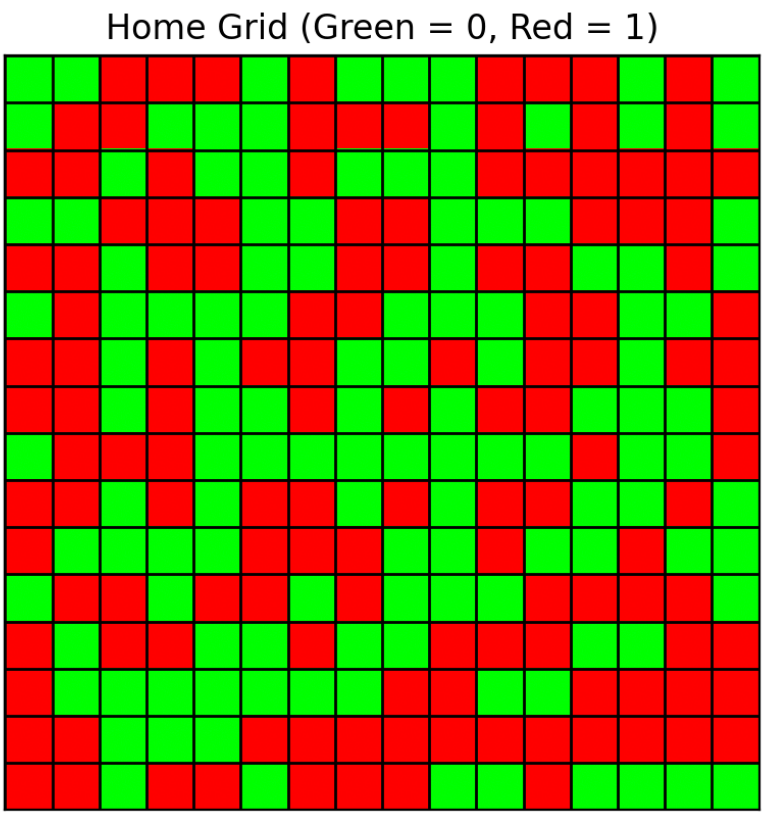
Example 4)

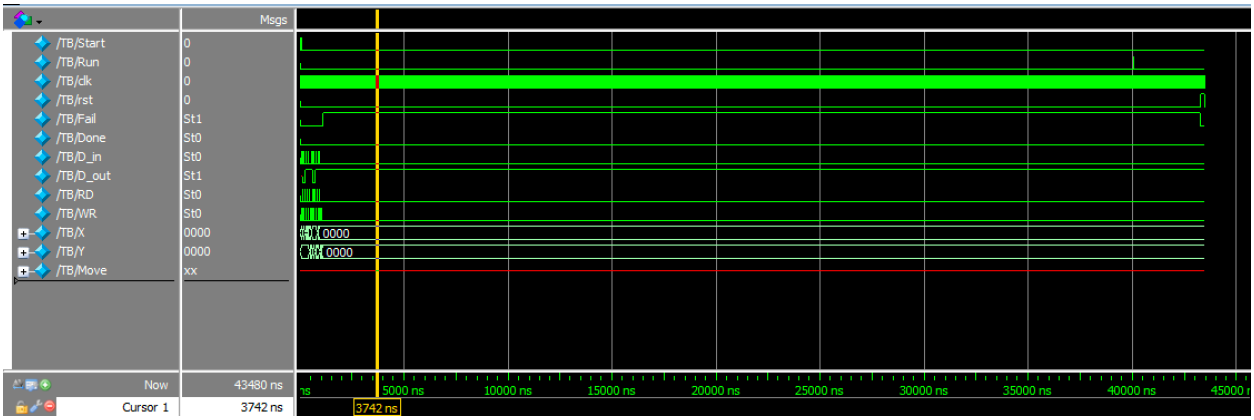


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right
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down
down
left
down
right
down
```



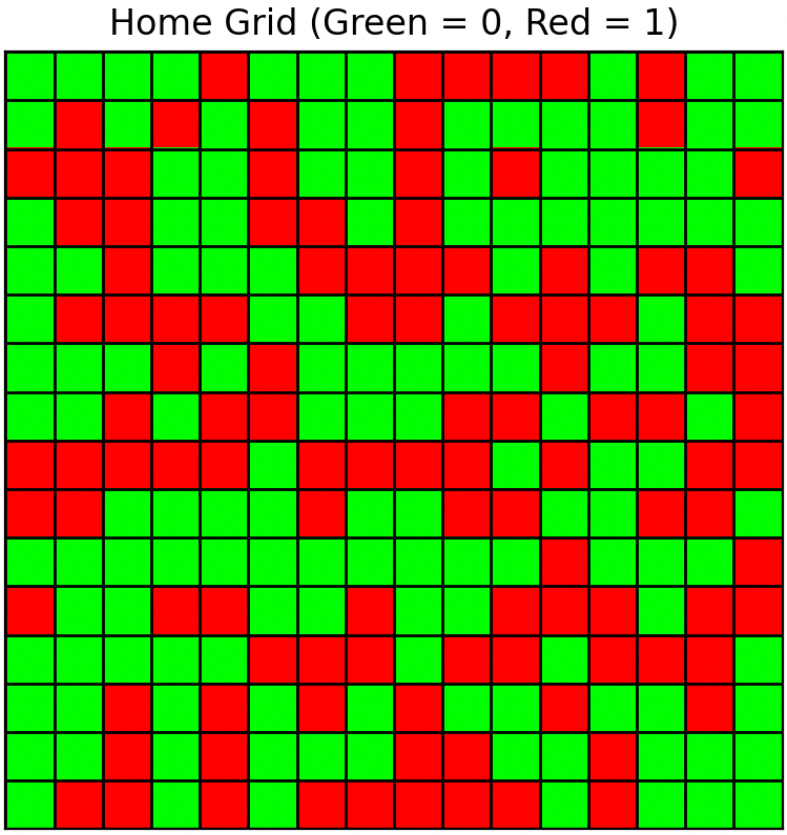
Example 5)

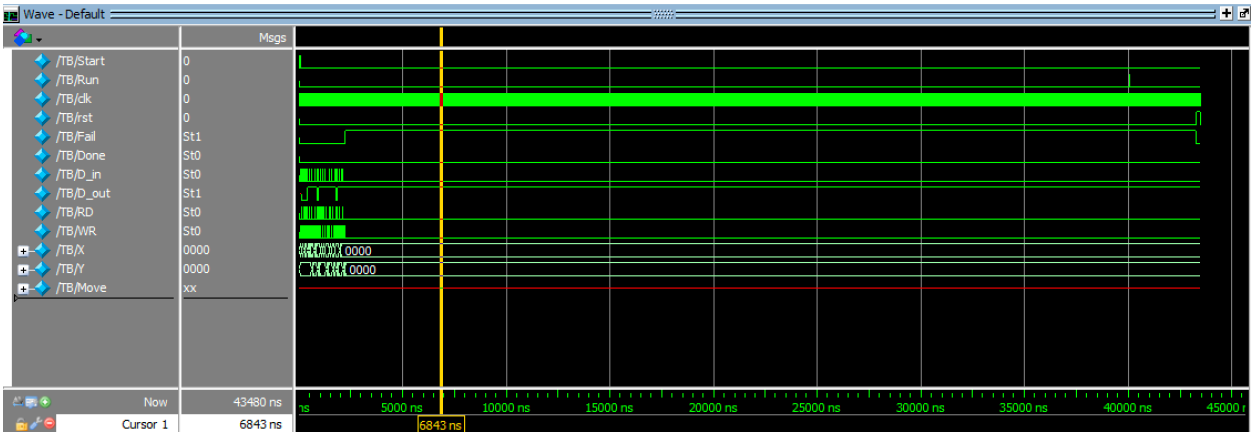



The fail = 1 so there’s no way!

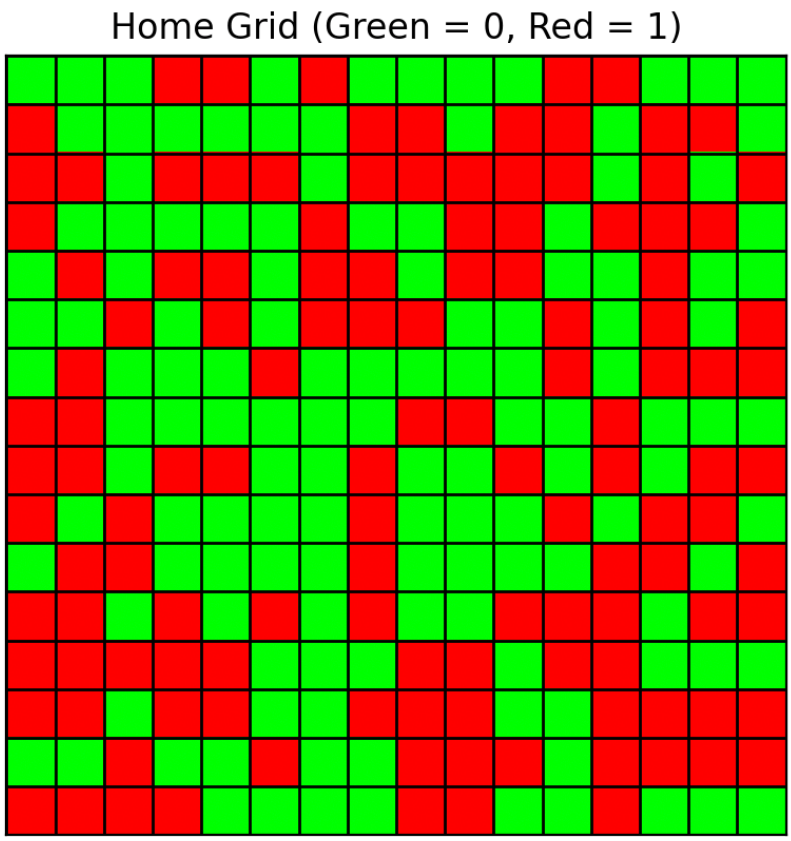
Example 6)

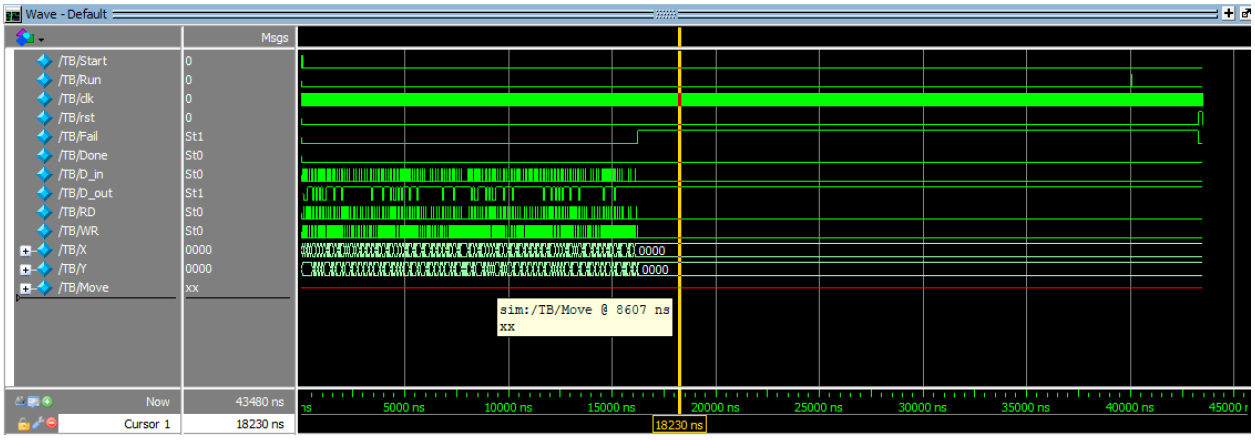



The fail = 1 so there's no way!

Example 7)




The fail = 1 so there's no way!